



The 24-Hour Build

Config-Driven Estimator & Owner Panel

A realistic slice of the work we do for paying clients. Read this document fully before you start — the constraints matter as much as the code.

TIME LIMIT

24 hours from the timestamp of your task email

REALISTIC EFFORT

8–11 focused hours

AI TOOLS

Fully allowed and expected

SUBMIT TO

careers@wantace.com

NEXT ROUND

Live interview on your own submission

01 Before You Begin — Read This Part

This is not a puzzle test and there is no hidden "correct" answer we are waiting for you to find. It is a compressed version of a real Wantace client build. We are evaluating how you think, what you choose to build, what you choose not to build, and whether you can defend every line of what you ship.

Three things that will save you

- **Scope beats ambition.** A smaller thing, finished and working, scores far higher than a larger thing left half-done. We would rather see three features that work than six that almost do.
- **The brief is deliberately incomplete**, exactly like a real client brief. Where it is silent, make a decision, write down why, and move on. Do not stall.
- **You may email us questions** at any point during the 24 hours at careers@wantace.com. We answer during working hours. Asking good questions early is a positive signal, not a weakness.

You are not expected to use all 24 hours. Sleep. We look at commit timestamps, and a submission built in steady blocks reads better than one built in a single panic at hour 22.

02 The Client Brief

Below is the email a client sent us. It is written the way owners actually write. This is all the information the client has given us so far.

From: Dale Whitmore, Northline Roofing & Exteriors (Columbus, OH)

Subject: the roof calculator thing we talked about

Hey — following up on our call. What I want is a thing on my website where a homeowner puts in a few details about their roof and gets a real cost estimate back, and I get their contact info. Right now people fill in my contact form saying "how much for a roof" and I have to call every one of them just to find out they wanted a shed done.

Big thing for me: I need to be able to change the prices myself. Material costs move constantly and last time I paid someone to build a site I had to email him every time I wanted a number changed and wait three days. Not doing that again. Same goes for the questions — I might want to add one about gutters later, or drop one that people keep getting confused by. I should be able to do that without calling anybody.

I also want to be able to see the leads that come in, obviously. Nothing fancy, I just need name, phone, and what they answered.

My guy Marcus who does the books will probably be in there too, so it can't be something where you need to be technical to use it. And whatever you build, it can't take the site down while I'm editing prices.

Let me know what you need from me.

03 What You Are Building

Two surfaces sharing one backend and one source of truth.

Surface 1 — The public estimator

- A multi-step estimator a homeowner can complete on a phone, ending in a cost estimate.
- **Every question, option, label, and rate is read from the database at runtime.** The front-end must not know what the questions are until it asks the server.
- Contact details are captured as part of the flow, and the lead is stored with the answers that produced the estimate.
- The estimate is calculated server-side from the stored configuration.

Surface 2 — The owner panel

- Behind a login. Basic auth is fine — this is not a security exercise, but an unprotected panel is a fail.
- The owner can edit rates and multipliers, edit labels, and turn a question on or off.
- The owner can view captured leads with their answers.
- Changes appear on the public estimator without a redeploy. Dale editing a price must never break the live tool for a homeowner mid-flow.
- It must be usable by someone who is not technical. Marcus the bookkeeper is your user here, not you.

Backend

- Your own API. Configuration and leads persisted in a real database — not a JSON file on disk, not browser storage.
- The calculation lives on the server. A visitor should not be able to read your pricing logic or tamper with the result from the browser.

Hard constraint — violating this is an automatic rejection

Do not hardcode any question, label, option, or rate in the front-end. If Dale might plausibly want to change it, it lives in the database and reaches the UI through your API. This single rule is the entire point of the task, and it is the first thing we check.

04 Seed Data

Use the following as your starting data. It is a real export from the client's current system, exactly as we received it. Treat it as production data — it is what it is, and part of your job is to deal with it as it stands.

Current configuration

```
{
  "config_version": 3,
  "business": { "name": "Northline Roofing & Exteriors", "region": "Columbus, OH", "currency": "USD" },
  "questions": [
    { "key": "roof_area", "label": "Roughly how big is your roof?", "type": "number",
      "unit": "sq ft", "required": true, "min": 300, "max": 12000, "active": true },

    { "key": "material", "label": "What material do you want?", "type": "select",
      "required": true, "active": true, "options": [
        { "value": "asphalt_3tab", "label": "Asphalt shingle - 3-tab", "rate_per_sqft": 4.25 },
        { "value": "asphalt_arch", "label": "Asphalt shingle - architectural", "rate_per_sqft": 5.90 },
        { "value": "metal_standing", "label": "Standing seam metal", "rate_per_sqft": 12.40 },
        { "value": "cedar_shake", "label": "Cedar shake", "rate_per_sqft": 11.10 } ] },

    { "key": "pitch", "label": "How steep is the roof?", "type": "select",
      "required": true, "active": true, "options": [
        { "value": "low", "label": "Low - you could walk on it", "multiplier": 1.0 },
        { "value": "medium", "label": "Medium", "multiplier": "1.12" },
        { "value": "steep", "label": "Steep - not walkable", "multiplier": 1.30 } ] },

    { "key": "layers", "label": "How many layers of old roofing are on there now?", "type": "select",
      "required": true, "active": true, "options": [
        { "value": "0", "label": "None - new build", "tear_off_per_sqft": 0 },
        { "value": "1", "label": "One layer", "tear_off_per_sqft": 1.15 },
        { "value": "2", "label": "Two or more layers", "tear_off_per_sqft": 2.05 } ] },

    { "key": "stories", "label": "How many stories is the house?", "type": "select",
      "required": true, "active": true, "options": [
        { "value": "1", "label": "Single storey", "multiplier": 1.0 },
        { "value": "2", "label": "Two storeys", "multiplier": 1.08 },
        { "value": "3", "label": "Three or more", "multiplier": 1.18 } ] }
  ],
  "modifiers": { "waste_factor": 0.10, "permit_flat_fee": 350, "range_spread_pct": 12 }
}
```

Existing leads

```
[
  { "id": "ld_1041", "captured_at": "2026-06-02T14:20:11Z", "config_version": 3,
    "name": "Ana Ruiz", "phone": "+1-614-555-0148", "email": "aruiz@example.com",
    "answers": { "roof_area": 2100, "material": "asphalt_arch", "pitch": "medium",
      "layers": "1", "stories": "2" },
    "estimate_low": 21480, "estimate_high": 27260 },

  { "id": "ld_0917", "captured_at": "2026-03-18T09:02:44Z", "config_version": 1,
    "name": "Bill Tanner", "phone": "+1-614-555-0192", "email": "btanner@example.com",
    "answers": { "roof_area": 1450, "material": "slate_natural", "pitch": "steep",
      "chimney_count": 2, "gutter_replace": "yes" },
    "estimate_low": 38900, "estimate_high": 44100 },

  { "id": "ld_1102", "captured_at": "2026-07-11T18:47:03Z", "config_version": 3,
    "name": "Priya Nair", "phone": "+1-614-555-0177", "email": "pnair@example.com",
    "answers": { "roof_area": 900, "material": "metal_standing", "pitch": "low",
      "layers": "0", "stories": "1" },
    "estimate_low": 12240, "estimate_high": 15530 } ]
```

The estimate figures in the seed leads are the client's historical numbers. Do not assume your formula must reproduce them exactly — define your own calculation and document it.

05 Ground Rules

AI tools	Fully allowed and expected. Claude, Cursor, Copilot, ChatGPT — whatever you use. We assume you will. You must be able to explain and modify every line you submit, live, in the next round.
Stack	Your choice, but we work in the MERN world. React or Next.js on the front, Node or Express on the back, MongoDB or Postgres underneath is the expected shape. Choosing something else is fine if you can justify it in your notes.
Libraries	Use whatever libraries you want, including UI kits. The configuration architecture and calculation layer must be your own work.
Design	Clean and credible is the bar. This is a tool a real business would put on its website. It does not need to be beautiful, but it cannot look like an unstyled form.
Questions	Email careers@wantace.com any time during the 24 hours. Answered during working hours. If we do not answer in time, decide and document.
Collaboration	Work alone. You may read documentation, Stack Overflow, and use AI. You may not have another person build any part of this.

06 What to Submit

Five things. All five are mandatory — a missing item is an automatic rejection regardless of code quality.

- 1 **A public Git repository** (GitHub or GitLab) with your real commit history. Commit as you go — do not squash 24 hours into one commit.
- 2 **A live deployed URL** for the public estimator, plus the owner panel URL and a test login. Vercel, Render, Railway, Netlify, or a VPS — all fine, all free-tier capable. It must still be up when we review it.
- 3 **README.md** — what you built, how to run it locally from a clean clone, environment variables needed, and the test credentials. Assume the reader has never seen your project.
- 4 **DECISIONS.md** — the most important file you will write. See below.
- 5 **AI_LOG.md** — a short, honest account of how you used AI on this build. See below.

DECISIONS.md — what we want in it

- Every assumption you made where the brief was silent, and why you chose that way.
- Your calculation formula, written out in plain language.
- What you deliberately did **not** build, and why that was the right call in 24 hours.
- Anything in the seed data or the brief that you found questionable, and how you handled it.
- The questions you would ask Dale before starting the real build.
- What you would do next if you had another week.

A page is plenty. We would rather have five sharp paragraphs than five pages of padding.

AI_LOG.md — what we want in it

Roughly half a page. Which tools you used, what you used them for, one specific instance where the AI produced something wrong or weak and what you did about it, and which parts of the codebase you wrote or substantially reworked yourself. **Honesty scores here. Claiming you wrote everything by hand does not.**

07 Optional Stretch Goals

Only touch these once everything above is finished, deployed, and working. **A half-finished stretch goal scores worse than no stretch goal at all** — shipping discipline is part of what we are measuring.

- Configuration version history, so Dale can see what changed and when.
- CSV export of captured leads from the owner panel.
- An outbound webhook that fires each new lead to a URL Dale sets in the panel.
- Letting the owner add an entirely new question, not just edit existing ones.
- Tests around the calculation layer.

08 How You Will Be Scored

We score every submission on the same seven dimensions, out of 100.

Dimension	Wt	What earns points
Correctness & logic	25	The calculation is right, including at the edges. Bad or unexpected input is handled deliberately. Nothing fails silently.
Agency & judgment	20	Quality of DECISIONS.md. Assumptions stated rather than guessed. Scope choices you can defend. Knowing what not to build.
Code quality	15	Readable, sensibly separated, no dead code, no 400-line file, named like a human wrote it.
UX & craft	15	Works properly on a phone. Loading and error states exist. Validation messages are useful. The owner panel is genuinely usable by a non-technical person.
Shipping & ops	10	Deployed, stable, and reachable. Clean environment config. A README a stranger can follow. Sane repo hygiene.
AI leverage & ownership	10	Evidence you steered the tool, rejected weak output, and verified results — rather than accepting whatever came back.
Effort ceiling	5	Stretch goals attempted and finished well. Unfinished ones score zero here.

09 Automatic Rejections

These are checked first, in under five minutes, before anything is scored. Any one of them ends the application:

- Questions, labels, options, or rates hardcoded in the front-end.
- No live deployed URL, or a URL that is down when we review it.
- The project does not run locally by following your own README.
- `DECISIONS.md` or `AI_LOG.md` missing.
- A single commit, or an entire history landing inside one short window.
- The owner panel is publicly accessible without a login.
- The core flow — complete the estimator, get an estimate, lead is stored — does not work.

10 Submission & What Happens Next

Deadline	24 hours from the timestamp of the email that delivered this document. Submitting early is fine and is a positive signal.
How to submit	Reply to your task email with: repository link, live estimator URL, owner panel URL, and test credentials. Nothing else needed — the rest lives in the repo.
If you run out of time	Submit what you have, working, with an honest note in DECISIONS.md about what is missing and why. A candid partial submission is respected. A broken submission presented as complete is not.
Round 2	A single interview. We will walk through your submission together, ask you to explain specific functions we pick, and give you a small live change to make to your own code with your usual AI tools available. Build accordingly — you will need to know your codebase.
Timeline	You will hear back within a few working days either way, with a reason.

One last thing

We know 24 hours is tight and we know you may have other commitments. We are not looking for a perfect product — we are looking for someone who can take an incomplete brief, make sound calls under time pressure, ship something real, and explain it honestly. Good luck.